

23CSE314 Compiler Design

Labsheet 2

```
C:\Users\exam\23219>java -cp .;antlr-4.13.2-complete.jar LexerDriver
Token: IF | Lexeme: if | Line: 1 | Column: 0
Token: ID | Lexeme: a | Line: 1 | Column: 3
Token: ASSIGN | Lexeme: = | Line: 1 | Column: 5
Token: INT | Lexeme: 10 | Line: 1 | Column: 7
Token: WHILE | Lexeme: while | Line: 2 | Column: 0
Token: ID | Lexeme: b | Line: 2 | Column: 6
Token: ASSIGN | Lexeme: = | Line: 2 | Column: 8
Token: ID | Lexeme: a | Line: 2 | Column: 10
Invalid token '+' at line 2, column 13
Token: INVALID | Lexeme: + | Line: 2 | Column: 12
Token: INT | Lexeme: 5 | Line: 2 | Column: 14
Token: ID | Lexeme: x | Line: 3 | Column: 0
Invalid token '@' at line 3, column 3
Token: INVALID | Lexeme: @ | Line: 3 | Column: 2
Token: ID | Lexeme: y | Line: 3 | Column: 4
```

Exercises

1. Design lexer rules for identifiers that:

- Start with a lowercase letter
- May contain digits
- Must end with a letter

Test with the following inputs:

a1b x9 ab12

a12b

```
C:\Users\exam\23219>java -cp .;antlr-4.13.2-complete.jar LexerDriver
Token: ID | Lexeme: a1b | Line: 1 | Column: 0
Invalid token 'x' at line 2, column 1
Token: INVALID | Lexeme: x | Line: 2 | Column: 0
Token: INT | Lexeme: 9 | Line: 2 | Column: 1
Token: ID | Lexeme: ab | Line: 3 | Column: 0
Token: INT | Lexeme: 12 | Line: 3 | Column: 2
Token: ID | Lexeme: a12b | Line: 4 | Column: 0
```

```

lexer grammar simple;
IF      : 'if' ;
ELSE    : 'else' ;
WHILE   : 'while' ;
INT      : [0-9]+ ;
ASSIGN   : '=' ;
ID : [a-z] [a-zA-Z0-9]* [a-zA-Z] ;
WS       : [ \t\r\n]+ -> skip ;
INVALID
: .
{
System.err.println(
"Invalid token '" + getText() +
"' at line " + getLine() +
", column " + getCharPositionInLine()
);
}
;

```

2. Write lexer rules to recognize:

- Decimal integers
- Hexadecimal integers (0x1A, 0xFF)(Ensure that 0x alone is rejected.)

Test with multi-line input.

```

lexer grammar simple;
IF      : 'if' ;
ELSE    : 'else' ;
WHILE   : 'while' ;
DECIMAL_INT : [0-9]+ ;
HEX_INT  : '0' [xX] [0-9a-fA-F]+ ;
ASSIGN   : '=' ;
WS       : [ \t\r\n]+ -> skip ;
INVALID
: .
{
System.err.println(
"Invalid token '" + getText() +
"' at line " + getLine() +
", column " + getCharPositionInLine()
);
}
;

```

```

C:\Users\exam\23219>java -cp .;antlr-4.13.2-complete.jar LexerDriver
Token: INT | Lexeme: 123 | Line: 1 | Column: 0
Token: HEX_INT | Lexeme: 0x1A | Line: 2 | Column: 0
Token: HEX_INT | Lexeme: 0xFF | Line: 3 | Column: 0
Token: INT | Lexeme: 0 | Line: 4 | Column: 0
Invalid token 'x' at line 4, column 2
Token: INVALID | Lexeme: x | Line: 4 | Column: 1
Token: INT | Lexeme: 456 | Line: 5 | Column: 0
Token: INT | Lexeme: 0 | Line: 6 | Column: 0
Invalid token 'X' at line 6, column 2
Token: INVALID | Lexeme: X | Line: 6 | Column: 1

```

3. Write appropriate ANTLR lexer rules (regular expressions) to recognize the following patterns. Test each rule with valid and invalid inputs.

- Floating point numbers with any number of digits before the decimal point and exactly 5 digits after the decimal point.
- Email id of the form <username>@am.amrita.edu .
- Vehicle registration of the form KL-XX YYYY.
- Domain names of the form <http://www.google.co.in>
<ftp://aaa.bbbbbb.com>
- Date format of the form: DD-MM-YYYY
- MAC Addresses of the form:00:1A:2B:3C:4D:5E

Input: KL-23 AB 1234

KL23AB1234

23-04-2006

00:1A:2B:3C:4D:5E

00:1A:2B:3C:4D:5E:

<https://www.google.com>

amrita@am.amrita.edu

```

lexer grammar simple;
ASSIGN : '=' ;
FLOAT_5DIGITS
: [0-9]+ '.' [0-9][0-9][0-9][0-9][0-9]
;S
EMAIL
: [a-zA-Z0-9._]+ '@am.amrita.edu'
;
VEHICLE_REG
: [A-Z][A-Z] '-' [0-9][0-9] ' '? [A-Z][A-Z] ' '? [0-9][0-9][0-9][0-9]
;
DATE
: [0-9][0-9] '-' [0-9][0-9] '-' [0-9][0-9][0-9][0-9]
;
DOMAIN
: (([a-zA-Z0-9] ([a-zA-Z0-9-]* [a-zA-Z0-9])? '.')+ [a-zA-Z][a-zA-Z-]+
;
MAC_ADDRESS
: [0-9A-Fa-f][0-9A-Fa-f] ':' [0-9A-Fa-f][0-9A-Fa-f] ':' [0-9A-Fa-f][0-9A-Fa-f]
': [0-9A-Fa-f][0-9A-Fa-f] ':' [0-9A-Fa-f][0-9A-Fa-f] ':' [0-9A-Fa-f][0-9A-Fa-f]
;
WS
: [ \t\r\n]+ -> skip
;
INVALID
: .
{
    System.err.println(
        "Invalid token '" + getText() +
        "' at line " + getLine() +
        ", column " + getCharPositionInLine()
    );
}
;

```

```

C:\Users\garla\desktop\Sem 6 Labsheets\Compiler Design\labsheet2>java -cp .;antlr4-4.9.2-complete.jar
LexerDriver
Token: VEHICLE_REG | Lexeme: KL-23 AB 1234 | Line: 1 | Column: 0
Invalid token 'K' at line 3, column 1
Token: INVALID | Lexeme: K | Line: 3 | Column: 0
Invalid token 'L' at line 3, column 2
Token: INVALID | Lexeme: L | Line: 3 | Column: 1
Invalid token '2' at line 3, column 3
Token: INVALID | Lexeme: 2 | Line: 3 | Column: 2
Invalid token '3' at line 3, column 4
Token: INVALID | Lexeme: 3 | Line: 3 | Column: 3
Invalid token 'A' at line 3, column 5
Token: INVALID | Lexeme: A | Line: 3 | Column: 4
Invalid token 'B' at line 3, column 6
Token: INVALID | Lexeme: B | Line: 3 | Column: 5
Invalid token '1' at line 3, column 7
Token: INVALID | Lexeme: 1 | Line: 3 | Column: 6
Invalid token '2' at line 3, column 8
Token: INVALID | Lexeme: 2 | Line: 3 | Column: 7
Invalid token '3' at line 3, column 9
Token: INVALID | Lexeme: 3 | Line: 3 | Column: 8
Invalid token '4' at line 3, column 10
Token: INVALID | Lexeme: 4 | Line: 3 | Column: 9
Token: DATE | Lexeme: 23-04-2006 | Line: 5 | Column: 0
Token: MAC_ADDRESS | Lexeme: 00:1A:2B:3C:4D:5E | Line: 7 | Column: 0
Token: MAC_ADDRESS | Lexeme: 00:1A:2B:3C:4D:5E | Line: 9 | Column: 0
Invalid token ':' at line 9, column 18
Token: INVALID | Lexeme: : | Line: 9 | Column: 17
Invalid token 'h' at line 11, column 1
Token: INVALID | Lexeme: h | Line: 11 | Column: 0
Invalid token 't' at line 11, column 2
Token: INVALID | Lexeme: t | Line: 11 | Column: 1
Invalid token 't' at line 11, column 3
Token: INVALID | Lexeme: t | Line: 11 | Column: 2
Invalid token 'p' at line 11, column 4
Token: INVALID | Lexeme: p | Line: 11 | Column: 3
Invalid token 's' at line 11, column 5
Token: INVALID | Lexeme: s | Line: 11 | Column: 4
Invalid token ':' at line 11, column 6
Token: INVALID | Lexeme: : | Line: 11 | Column: 5
Invalid token '/' at line 11, column 7
Token: INVALID | Lexeme: / | Line: 11 | Column: 6
Invalid token '/' at line 11, column 8
Token: INVALID | Lexeme: / | Line: 11 | Column: 7
Token: DOMAIN | Lexeme: www.google.co | Line: 11 | Column: 8
Invalid token 'm' at line 11, column 22
Token: INVALID | Lexeme: m | Line: 11 | Column: 21
Token: EMAIL | Lexeme: amrita@am.amrita.edu | Line: 13 | Column: 0

```

4. Modify the driver program to count:

- Number of keywords
- Number of identifiers
- Number of literals
- Number of invalid tokens

```
C:\Users\garla\desktop\Sem 6 Labsheets\Compiler Design\labsheet2>java -cp .;antlr4-4.9.2-complete.jar LexerDriver
line 1:0 token recognition error at: 'KL-2'
line 1:4 token recognition error at: '3 '
line 1:6 token recognition error at: 'AB '
line 1:9 token recognition error at: '1234\r'
line 3:0 token recognition error at: 'KL23AB1234\r'
Token: DATE | Lexeme: 23-04-2006 | Line: 5 | Column: 0
Token: MAC_ADDRESS | Lexeme: 00:1A:2B:3C:4D:5E | Line: 7 | Column: 0
Token: MAC_ADDRESS | Lexeme: 00:1A:2B:3C:4D:5E | Line: 9 | Column: 0
line 9:17 token recognition error at: ':'
line 11:0 token recognition error at: 'https:'
line 11:6 token recognition error at: '/'
line 11:7 token recognition error at: '/'
line 11:8 token recognition error at: 'www.google.com\r'
Token: EMAIL | Lexeme: amrita@am.amrita.edu | Line: 13 | Column: 0

=== Summary ===
Keywords: 0
Identifiers: 0
Literals: 4
Invalid tokens: 0
```